

SEMIYIELD AI

Predict Yield.
Explain Failure.
Optimize Process.

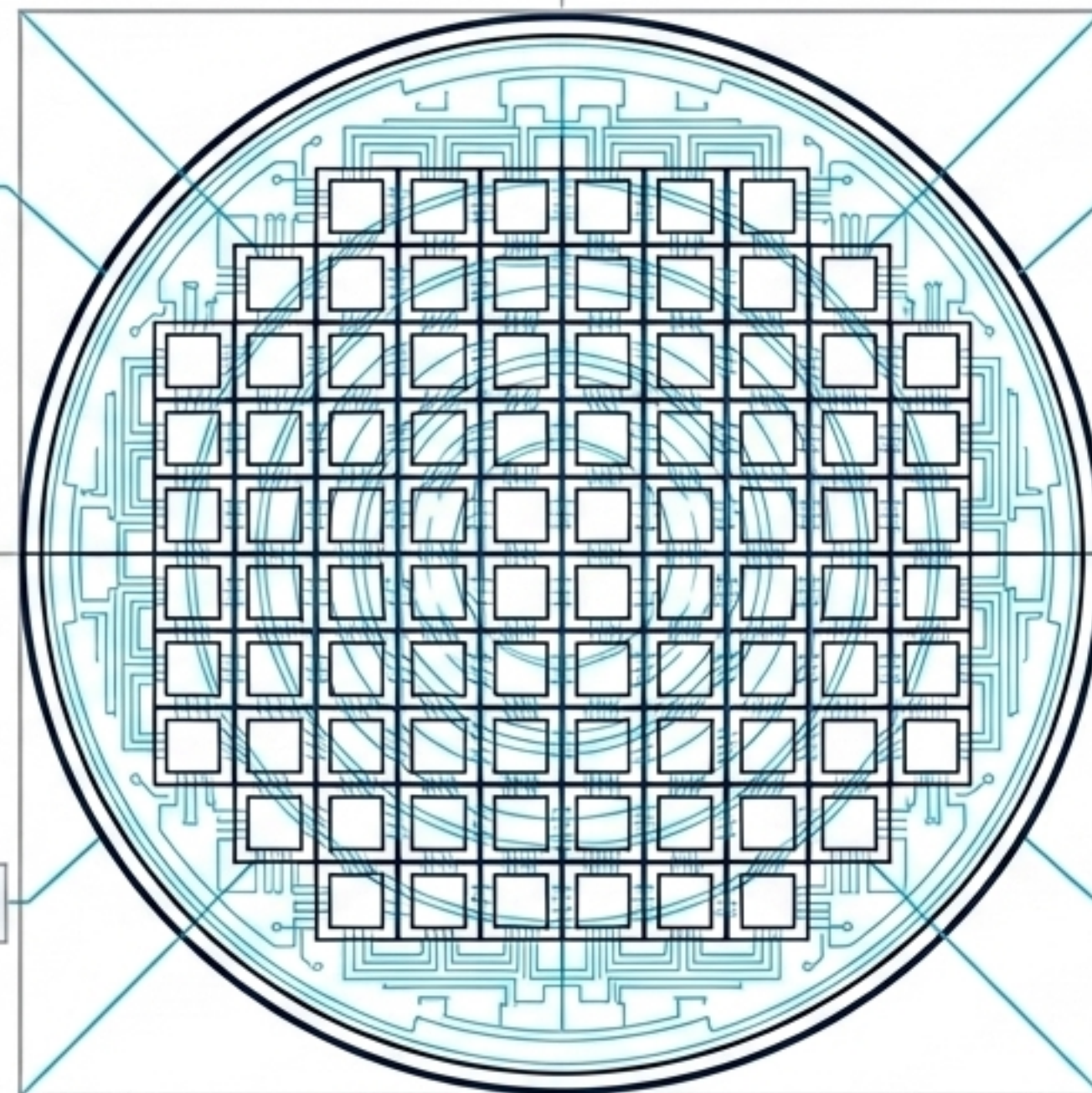
AI-powered semiconductor yield intelligence for predictive manufacturing.

YIELD_INTELLIGENCE

ROOT_CAUSE_AI

PREDICTIVE_ANALYTICS

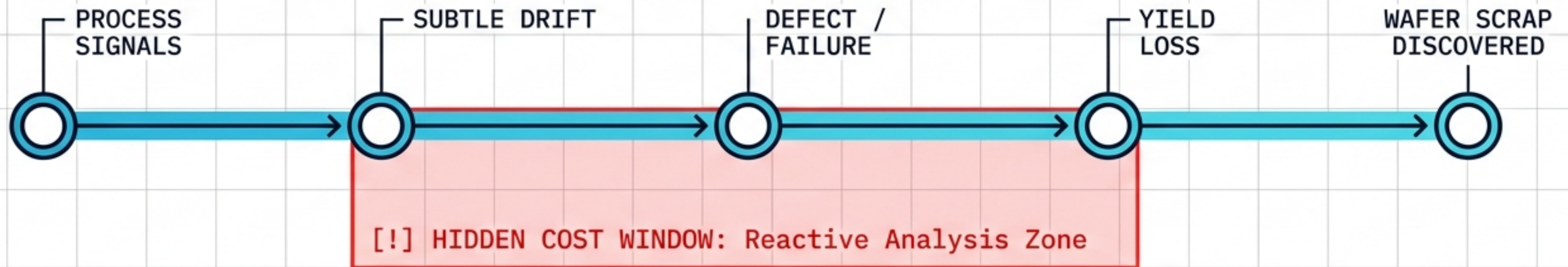
RECIPE_OPTIMIZATION



Presented by The Fourhorsemen
IBM Bob AI Hackathon

SEMICONDUCTOR YIELD LOSS IS OFTEN DISCOVERED TOO LATE.

In advanced nodes, a 1% yield loss costs \$50M–\$100M per month.
Identifying failure precursors across 500+ sensors takes weeks of manual analysis.



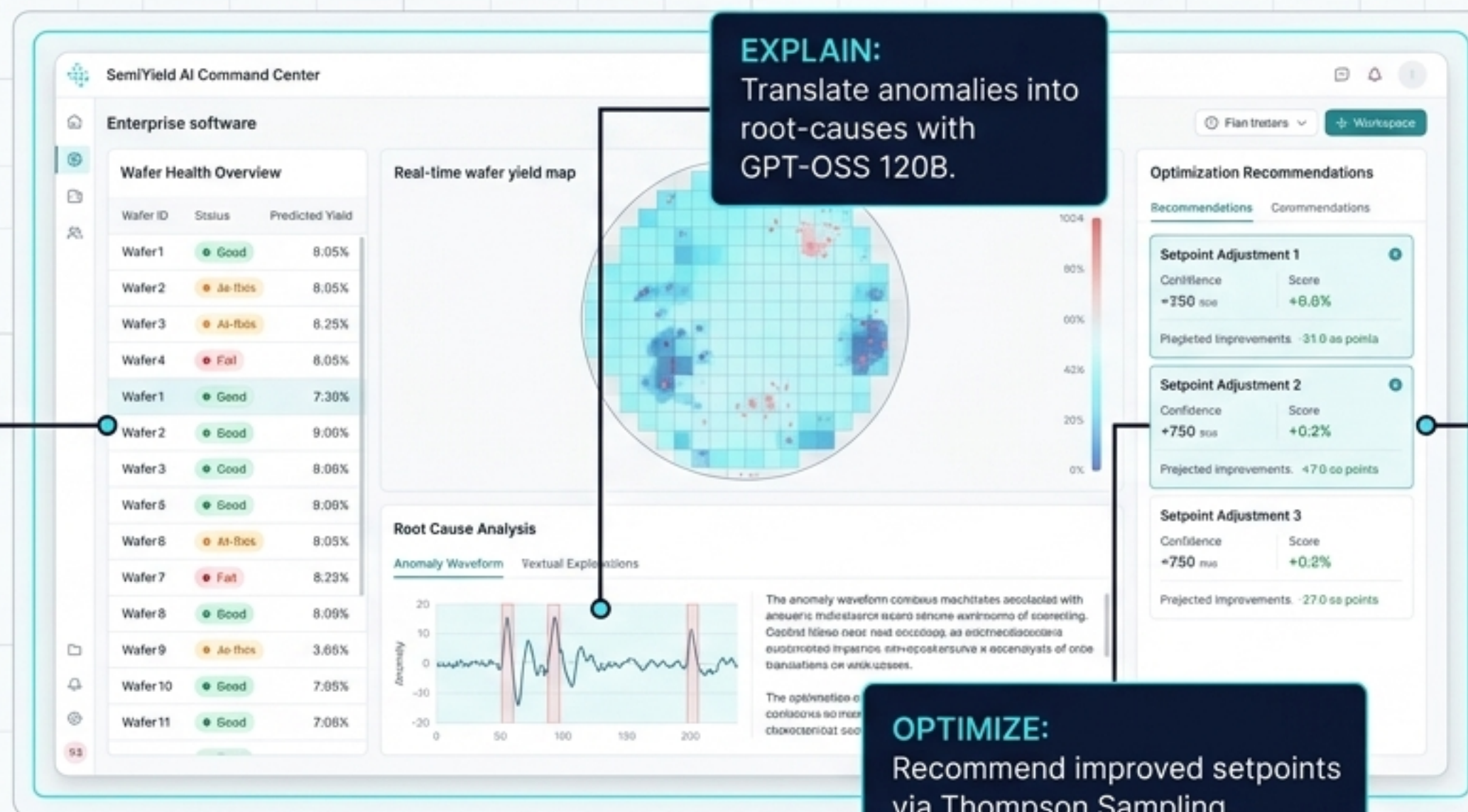
Reactive analysis is not enough when every failed wafer has a cost.

FROM PROCESS SIGNALS TO ENGINEERING ACTION

A closed-loop AI platform built for the fab command center.



PREDICT:
Identify at-risk wafers
via ONNX engine.



OPERATE:
Unify intelligence in a
single workspace.

Don't just predict failure. Prevent it.

SENSE → **PREDICT** → EXPLAIN → OPTIMIZE

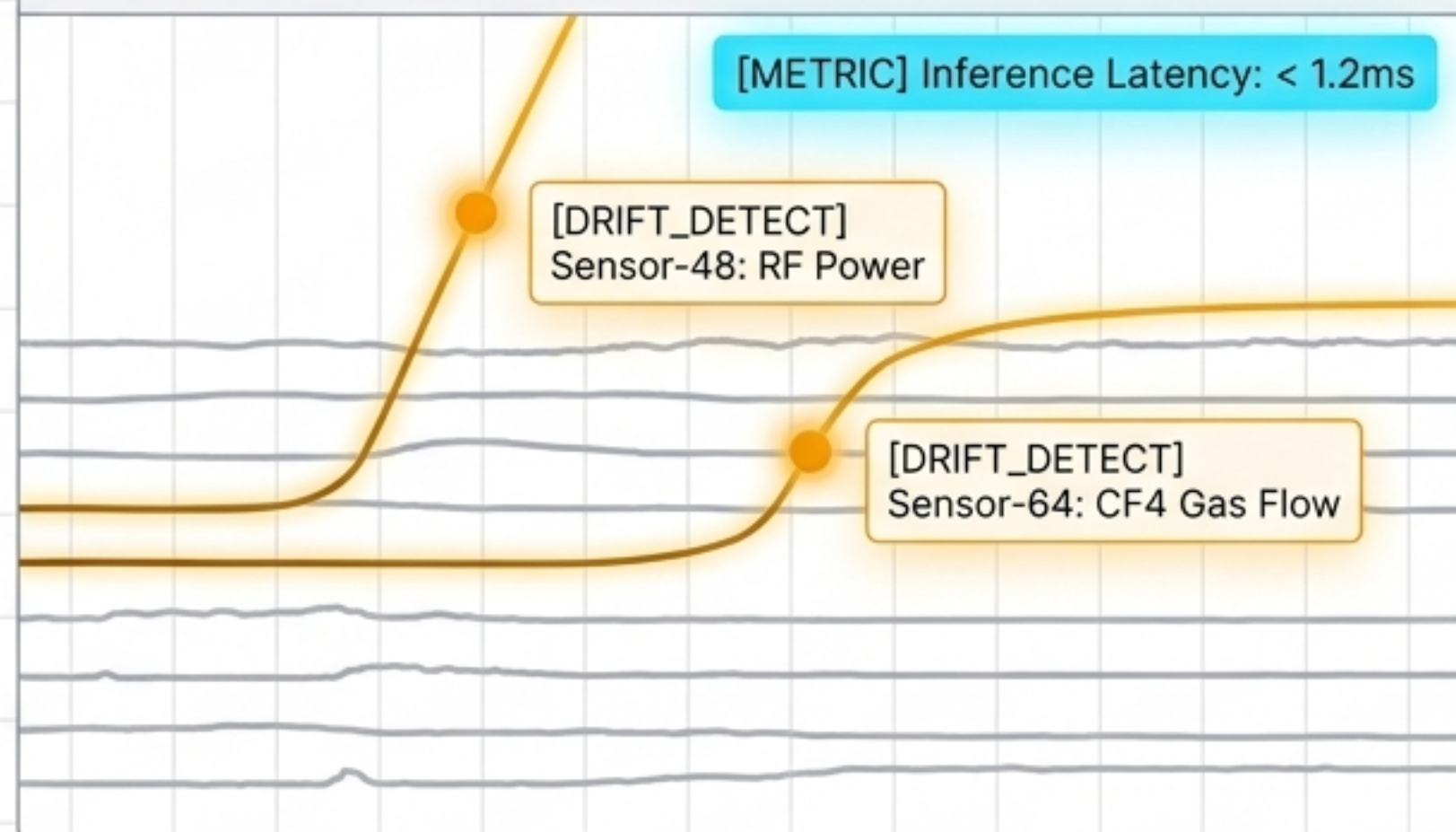
PREDICT: SUB-MILLISECOND YIELD INFERENCE.

An extreme-performance predictive core detecting anomalies before wafers leave the chamber.

INGESTION & ENGINE

DATASET:	UCI SECOM Benchmark
VOLUME:	1,567 Wafers × 590 Sensors
IMBALANCE:	1:14 Failure Class Ratio
MODEL:	Tuned CatBoost + Tree Ensemble
RUNTIME:	ONNX Runtime Engine

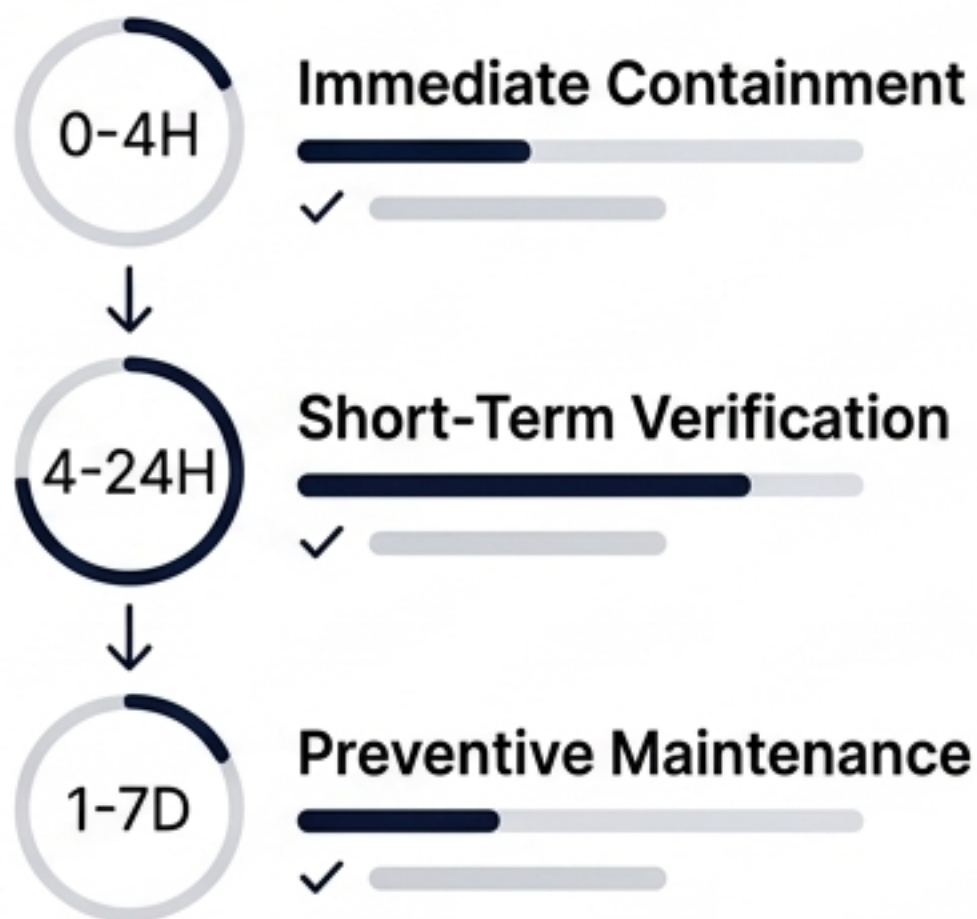
REAL-TIME SIGNAL ANALYSIS



EXPLAIN: PRESCRIPTIVE ROOT-CAUSE COPILOT.

Powered by Groq Cloud GPT-OSS 120B MoE, translating sensor anomalies into immediate engineering directives.

3-Phase Action Roadmap



Financial Impact

STATUS: Scrap Cost Prevented

\$15,000+

METRIC: Yield Percentage Uplift

+0.4%

RISK: Cycle Time Evaluation

Low Impact

SPC & Recipe Directives

TRIGGER: Western Electric & Nelson SPC Rules violated.

ADJUST: Recipe Parameter Offsets (RF Power, CF4 Flow, ESC Temp).

- > SPC Rule 2: 7 points > centerline (Western Electric)
- > SPC Rule 3: 3 of 4 points > 2σ (Nelson)

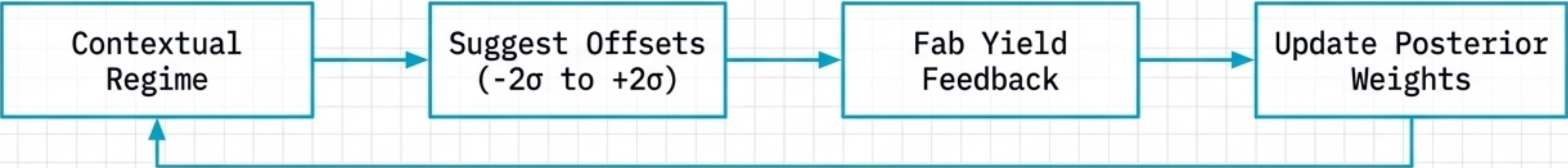
\$ Set RF_Power: 750W → 735W (Offset: -15W)
\$ Set CF4_Flow: 200sccm → 195sccm (Offset: -5sccm)
\$ Set ESC_Temp: 60C → 58C (Offset: -2C)

SENSE → PREDICT → EXPLAIN → OPTIMIZE

OPTIMIZE: ADAPTIVE CLOSED-LOOP CONTROL.

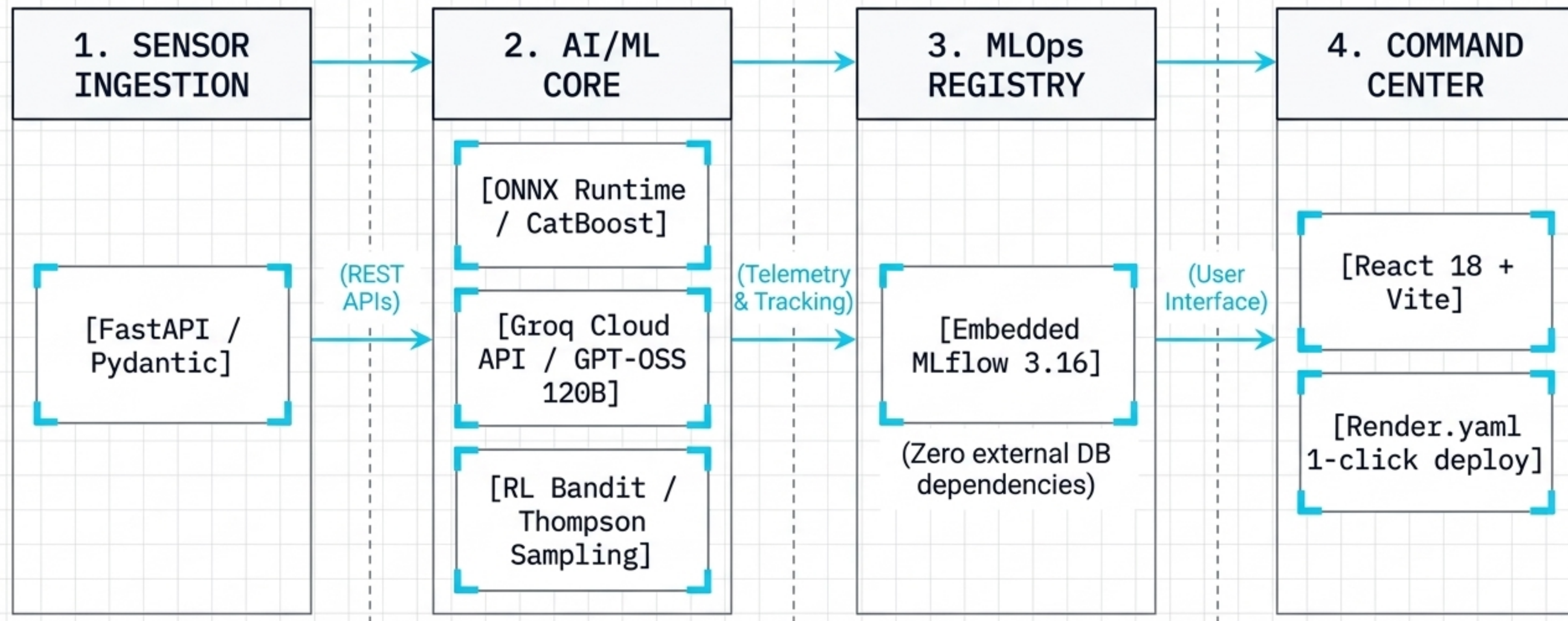
A multi-action Contextual Bandit continuously updating recipe setpoints based on live yield feedback.

Dimension	Static ML Models	SemiYield RL Bandit
Learning Mechanism	Requires heavy full-model retraining	Continuous online learning
Drift Handling	High risk of catastrophic forgetting	Adapts via fab yield feedback
Exploration	Fixed, rigid setpoints	Thompson Sampling (Beta-Bernoulli priors)
Regime Context	Single monolithic state	8 clustered operational regimes
Action Space	Predicts Pass/Fail	5 Process Params × 5 Offset Bins



OPERATE: ENTERPRISE-READY ARCHITECTURE.

Fully decoupled REST API, embedded MLflow tracking, and single-command Render deployment.



THE FOUR HORSEMEN

Tirth (lead)

frontend engineering
+ ml integration

- frontend architecture and ui
- model execution pipeline
- model serving integration
- product experience

Manan

ml engineering +
model evaluation

- model building
- model training
- model evaluation
- ml pipeline

Nil

devops + data
engineering

- devops / deployment
- infrastructure
- data engineering
- data support for model development

Bhakti

frontend engineering

- frontend development
- ui/ux implementation
- product interface